

ITAI 1371 Introduction to Machine Learning

Professor Viswanatha Rao (Vishwa)

Assignment: Lab 06 Exploratory Data Analysis

Group Member Contributions

Monepha Mitchell

My individual contribution to this lab included completing and reviewing both required machine-learning tasks. I trained the Linear Regression model to predict passenger fares using age and passenger class, generated predictions, and evaluated the model with mean squared error and root mean squared error. I also prepared the classification data, trained the Logistic Regression model to predict passenger survival, and evaluated its performance with accuracy. In addition, I added comments to explain the main coding steps and answered the knowledge-check questions in my own words. One important takeaway from this lab was that the evaluation metric must match the type of problem: regression metrics measure numerical prediction errors, while classification metrics measure how correctly the model assigns class labels.

Ved Dalwadi

For module 7, I ran the code on the confusion matrix and classification report to understand the output of precision and recall scores for predicting titanic survival. I took these 2 terms and turned them into real world examples to help me and my teammates better understand each of the concepts. For precision, I used self driving cars and airport security; for recall, these examples helped break down the terms given to us. I also reviewed the cross validation method to help me understand why this is better than just a train test split

Nigel Bontif,

For Module 06, I contributed by working through the Regression and Classification lab notebook and helping complete the model-building sections. I worked on the Linear Regression part for predicting passenger fare and the Logistic Regression part for predicting survival. I also helped clean up the code structure so the cells would run correctly, reviewed the model outputs, and completed the knowledge check answers. My main focus was making sure I understood and explained the difference between regression as predicting a number and classification as predicting a category.